

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

April-may 2012

EC 302 Integrated Circuits & Applications

Date: 1.5.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) In the given circuit of Fig. (1) Find out the output voltage. [3]
(b) Explain ideal voltage transfer curve for op-amp and also explain the ideal characteristic of an op-amp. [4]
- Q - 2 (a) Explain op-amp as Differentiator. Discuss limitation of basic Differentiator. How it is overcome? [6]
(b) For the inverting amplifier of Fig.2, determine the maximum possible output offset voltage due to :
(1) input offset voltage V_{io} (2) input bias current I_B [5]
What value of offset minimizing resistor (R_{om}) is needed to reduce the effect of input bias current I_B ? (consider $V_{io\max}=6\text{ mv}$ & $I_{B\max}=500\text{ na}$)
(c) Define (i)input offset voltage (ii) Differential input resistance (iii)CMRR [3]

OR

- Q - 2 (a) Design a differentiator to differentiate an input signal that varies in frequency from 10 Hz to 1kHz and draw its output waveform, if a sine wave of 1 V peak at 1000Hz is applied to the differentiator. [6]
(b) Find out input resistance for voltage shunt and output resistance for voltage series feedback amplifier. [5]
(c) Write short note on voltage to current converter with grounded load. [3]
- Q - 3 (a) For Square wave generator prove that $f_o = \frac{1}{2RC * \ln\left(\frac{2R_1 + R_2}{R_2}\right)}$ [5]
(b) Explain differential input and differential output amplifier. If it is used as a preamplifier and requires a differential output of at least 3.7V, determine the gain of the circuit input $V_{in}=100\text{mV}$. [5]
(c) The monostable multivibrator using 555 is to be used as a divide by 2 networks. The frequency of input trigger is 2kHz.if the value of $c=0.01\mu\text{F}$, What should be the value of R_A ? Assume suitable data. [4]

OR

- Q - 3 (a) How 555 timer act as an astable multi-vibrator? [5]
(b) Explain with circuit diagram Absolute value output circuit. [5]
(c) Describe the working peak detector circuit. [4]

SECTION – II

- Q - 4 (a) Sketch the schematic diagram of LM380 audio power amplifier. [3]
 (b) Draw and Explain Block diagram of Phase Locked Loop. [4]

- Q - 5 (a) Explain basic switching regulator with circuit diagram. Compare fixed regulator with Switching regulator. [8]
 (b) Using LM317, Design an adjustable voltage regulator to satisfy the following specifications: output voltage $V_0=5\text{V}$ to 12V , output current $I_0=1\text{A}$, $I_{\text{ADJ}}=100\mu\text{A}$, $R_1=240\Omega$ and $V_{\text{REF}}=1.25\text{V}$ [3]
 (c) Using the 7805C voltage regulator, design a current source that will deliver 250mA current to the 48Ω , 10-W load. Draw the complete schematic diagram of the circuit. [3]

OR

- Q - 5 (a) Design the circuit to realize bilinear transfer function with a zero at $f_z=830\text{Hz}$, a pole at $f_p=13\text{kHz}$ and high frequency gain of 23dB . [6]
 (b) Derive equation for magnitude and phase angle of Bilinear transfer function. [8]
 Q - 6 (a) Derive element scaling equation for L_{new} , C_{new} & K_{new} in terms of K_m , K_f & the old elements. [7]
 (b) Design a circuit to realize the transfer function of the Bode plot given in Fig.3. Use inverting configuration. [7]

OR

- Q - 6 (a) Explain Sallen-Key circuits. [7]
 (b) For the following given specification, design butter worth low-pass filter [7]
 $\alpha_{\text{max}}=0.5\text{dB}$, $\alpha_{\text{min}}=20\text{dB}$, $\omega_p=1000\text{rad/s}$, $\omega_s=2000\text{rad/s}$

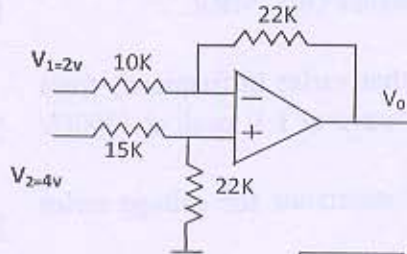


Fig 1

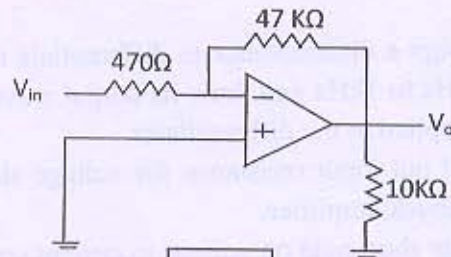


Fig 2

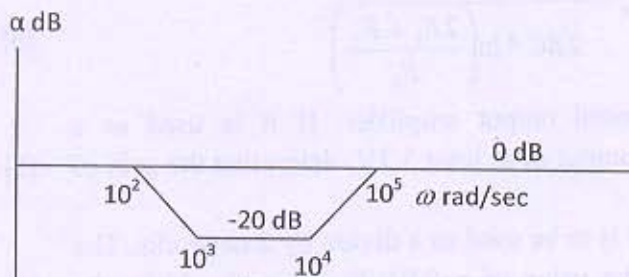


Fig 3